Mruthulan Senthil Nathan

SOFTWARE ENGINEERING INTERN | FULL-STACK DEVELOPMENT

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Available for a year-long internship in 2027/2028

EDUCATION

Diploma in Information Technology, Singapore Polytechnic | Apr 2025 – May 2028 (Expected)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL
Web: React, Next.js, Node.js, Express, FastAPI, REST APIs, HTML/CSS
Databases: PostgreSQL, MySQL, SQLite
Tools: Git, GitHub, pytest, Render

SELECTED PROJECTS

SignalBridge

Youth support platform with consent and human handoff | Team of four, Singapore Children's Society brief
Champion, Dell InnovateDash 2026 | Featured by Singapore Polytechnic

- Built the youth chat, sign-in, dashboard and the consent screen where the young person reviews the note before it is shared.
- Built FastAPI chat routes and the Discord intake integration, with pytest tests for the intake (Next.js, FastAPI, PostgreSQL).

Boss Breaker

Solo full-stack wellness game | Backend module coursework | github.com/mru34/bedca2

- Built a nine-page frontend, an Express API and a nine-table MySQL schema for challenges, boss raids, the shop and inventory.
- Calculated points and boss damage on the server so the browser cannot cheat; secured accounts with bcrypt password hashing and JWT-protected routes.

Better Call Bhai

Booking site for a Singapore barbershop
Deployed pilot with the shop

- Built the booking flow: customers pick a service, a day and an open slot and confirm on the site, with their booking details one tap away on WhatsApp.
- Built a Node.js/Express API on SQLite with server-side rules (no past dates; each slot booked once) and an admin page for the shop; deployed on Render.

MEANT

Assistive communication (AAC) app that suggests replies on-device; the user picks every word
2nd Runner-Up, polytechnic category (S$3,000), Dell InnovateFest 2026 | Featured in Tamil Murasu

- Designed the UI and interaction flows: colour-coded symbol boards, search across the whole vocabulary when a prediction is wrong, and setup that stays out of the conversation.
- Fixed a stuck mic, a silent send and a black camera before the live demo; worked on the Singaporean text-to-speech voice with a teammate and presented the build.

AWARDS, EXPERIENCE AND LEADERSHIP

- **Champion, Autodesk Singapore Hackathon 2026** – KnowCad, a retrieval-based AI assistant for customer service. Built and delivered the winning presentation; Autodesk engineers led the code.
- **Retail Associate, Watsons Singapore** | Mar – Apr 2025 – Served customers, handled payments and self-checkout, and replenished stock.
- **Secretary, Subcommittee, Youth Harmony Chapter** | Jun 2026 – Present – Coordinated the subcommittee's communication and followed up on actions for student activities.
- **Class Chairman, Singapore Polytechnic** | 2025 – Led class communication.
- **SP ACER, Singapore Polytechnic** | May 2025 – Present – Supported outreach, including Open House 2026 and First Steps with SP.